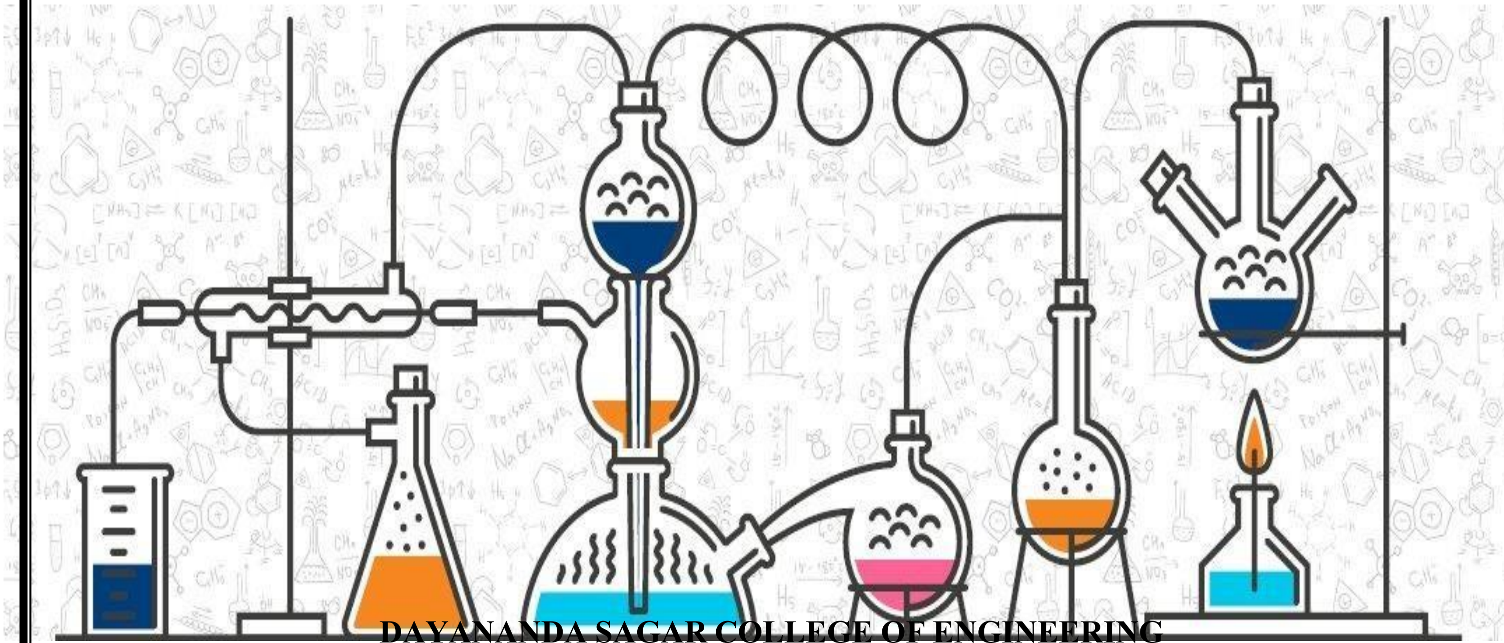
II SEMESTER (1BCHES102/202)

Smart Materials for Emerging Technology - CS Stream

Module – 3
Memory Devices and Polymers



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Definition of Memory Devices

A memory device is a hardware component used to store data and instructions either temporarily or permanently. They are essential elements in modern electronic systems such as computers, smartphones, tablets, cameras, and embedded devices. Memory devices enable storage and retrieval of digital information in the form of binary code ("0" and "1"), which can represent data and program instructions.

Semiconductors are materials with electrical conductivity between conductors and insulators. Their conductivity can be controlled by temperature and dopants.

Inorganic Semiconductors:

Inorganic semiconductors are materials composed of inorganic elements or compounds that exhibit semiconducting properties due to the presence of a moderate band gap between the valence and conduction bands. These materials typically possess a crystalline lattice structure in which charge transport occurs through the movement of electrons and holes within extended energy bands.

Inorganic semiconductors conduct electricity when energy (thermal, electrical, or optical) excites electrons from the valence band to the conduction band. Their electrical conductivity can be precisely controlled through doping with suitable impurities, enabling the formation of n-type and p-type materials. Similar to organic semiconductors, inorganic semiconductors can operate in different conductivity states (ON and OFF) under an applied electric field, forming the basis of modern electronic devices.

Common examples include Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs), and Zinc Oxide (ZnO).

These materials are widely used in high-performance and durable electronic devices such as integrated circuits, microprocessors, solar cells, LEDs, laser diodes, and power electronics due to their high charge carrier mobility, thermal stability, and mechanical robustness.

Inorganic semiconductors offer high performance and stability.

Organic Semiconductors:

Organic semiconductors are materials made primarily of carbon-based molecules or polymers that have semiconducting properties. Organic semiconductors consist of π-bond and lone pair molecules or polymers that enable charge transport through electron delocalization within conjugated systems of carbon atoms. Organic semiconductors store data based on different electrical conductivity states (ON and OFF bistable states) in response to an applied electric field.

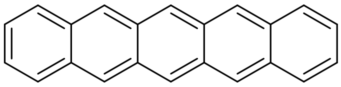
These materials can be processed into flexible, lightweight, and cost-effective electronic devices such as organic light-emitting diodes (OLEDs), organic photovoltaics (solar cells), and organic field-effect transistors (OFETs).

Types of organic memory devices: Bases on materials which can exhibit bistable states 1) Organic molecules: Pentacene, perfluoropentacene 2) Polymeric materials: Polyimide 3) Organic-Inorganic hybrid materials: Composite of a polystyrene, gold nanoparticles, and 8 hydroxyquinoline.

p-type Semiconductor Memory devices. (pentacene)

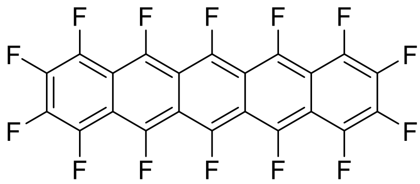
An Organic molecule with π conjugated system and possesses holes as major charge carrier is called p-type semiconductor.

Example: **Pentacene**: Pentacene is a linear aromatic hydrocarbon formed by the fusion of five benzene rings. The extended π-system allows the continuous delocalization of π-electrons and there is a lateral over lapping of pi-electrons between the molecules.



pentacene

The n-type organic semiconducting material (Perfluoropentacene)



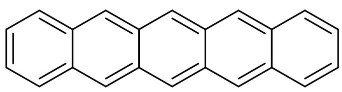
It is based on a π-conjugated pentacene backbone in which all hydrogen atoms are replaced by fluorine atoms. The strongly electron-withdrawing nature of fluorine makes perfluoropentacene highly electron-deficient and stable, so it accepts and transports electrons as the majority charge carriers. In memory devices, perfluoropentacene is typically sandwiched between two electrodes: an n-type electrode that donates electrons and a p-type electrode that accepts them. When a voltage is applied, electrons inject from the n-type electrode into the perfluoropentacene layer, changing its electrical properties such as resistance or capacitance. This change can be used to store a bit of information, for example a high resistance representing "0" and a low resistance representing "1." The stored state is read by applying a smaller read voltage and measuring the resulting current, which reflects the resistance state of the perfluoropentacene layer.

Differences between organic and inorganic memory devices

Feature / Property	Organic Memory Devices	Inorganic Memory Devices
Active material	π and lone pair-conjugated organic molecules or polymers (e.g., pentacene, PFP, polyimide)	Silicon-based materials or metal oxides (e.g., SiO_2, TiO_2, $Ge_2Sb_2Te_5$)
Device fabrication	Solution processing, printing, spin-coating; low-temperature, flexible substrates.	High-temperature vacuum deposition and photolithography on rigid wafers
Substrate & flexibility	Flexible (plastic, paper, etc.), lightweight, bendable	Rigid (silicon, glass), not easily flexible
Cost and scalability	Low-cost, highly scalable to large-area and roll-to-roll production	Higher cost, limited scalability to flexible or large-area systems
Environmental impact	Organic, less toxic, more eco-friendly	Metal-based, potential toxic byproducts and higher environmental footprint

Construction, working and advantages of pentacene semiconductor chip.

Pentacene is an organic electronic device made using pentacene as the semiconducting layer. It consists of five linearly fused benzene rings, forming a planar, conjugated π-system. Pentacene is a p-type semiconductor, meaning it has a deficiency of electrons and primarily conducts via holes



pentacene

Construction:

Pentacene semiconductor chips are comprised of the following layers and components:

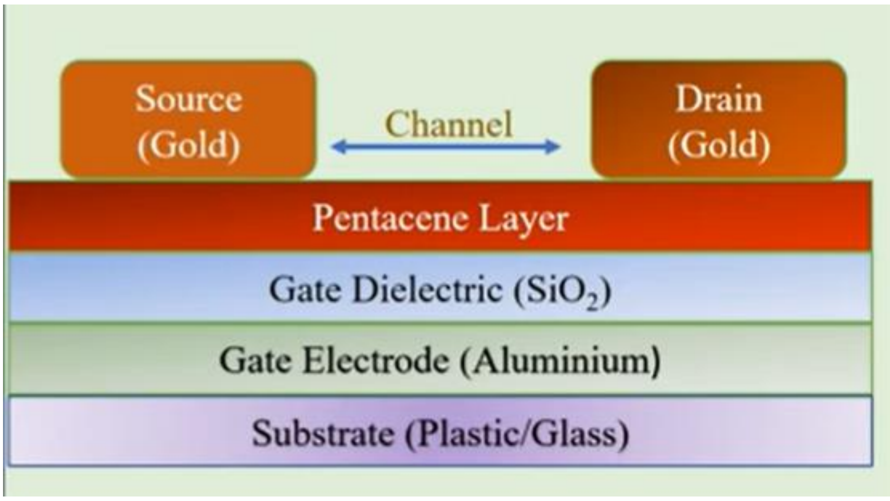
Substrate: Usually flexible plastics, glass or silicon wafer can be used.

Gate electrode: A thin metal layer like Au or Al is deposited on the substrate. This layer act as gate electrode.

Gate dielectric layer (Insulating): A thin insulating material such as SiO2 is deposited over the gate to act as insulator for the gate electrode.

Pentacene Layer: Pentacene is p-type organic semiconductor, is deposited to form thin ordered film. The layer is usually deposited by evaporation method.

Source and drain electrodes: Two metal electrodes for the source (where charge enters) and drain (where charge leaves) are deposited on the top of the pentacene layer to create the current carrying path.



Working:

- When a voltage is applied to the gate electrode, an electric field is created across the dielectric layer. The electric field induces charge carriers (holes) in the pentacene layer, forming a conductive channel between source and drain electrodes.
- This electric field modulates the conductivity of the pentacene layer between the source and drain electrodes.
- Charge carriers (holes, since pentacene is a p-type semiconductor) move through the pentacene layer, allowing current to flow from source to drain when the device is in the 'on' state.
- The flow of electrons causes a change in the electrical properties of the pentacene layer, such as a change in its resistance or capacitance. High resistance = ‘0’ (OFF) and Low resistance = ‘1’ (ON). This change can be used to represent a bit of information, such as a 0 or 1.
- The state of the memory device can be read by applying a smaller voltage across the two electrodes and measuring the resulting current. The current will be higher or lower depending on the state of the pentacene layer.

Advantages:

• Low fabrication cost

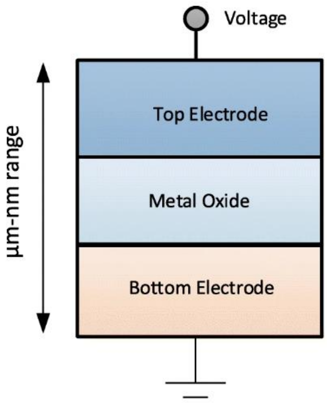
• Lightweight and flexible

- Compatible with large-area processing
- Low operating voltage and power consumption
- Suitable for flexible electronics and displays

Resistive RAM (ReRAM) Materials:

Introduction:

Resistive RAM (ReRAM or RRAM) materials, typically structured as a metal-insulator-metal stack, rely on creating conductive filaments through a dielectric layer. Common materials include binary transition metal oxides (TiO2), solid-state electrolytes (GeS, Cu2S), and 2D materials (graphene) that offer high scalability and low-power operation



TiO_2 (titanium dioxide) is widely used as the resistive switching layer in ReRAM devices due to its excellent electrical insulating properties, chemical stability, and compatibility. TiO_2 exhibits reliable bipolar and unipolar resistive switching behavior, driven mainly by the formation and rupture of conductive filaments composed of oxygen vacancies or titanium interstitials. This makes TiO_2 an ideal candidate for low-power, high-speed, and scalable memory application.

Synthesis of TiO_2-RAM nanomaterial by sol-gel method

Sol-gel process: Sol-gel process has been used in the synthesis of nanoparticles of metal oxides and temperature sensitive organic-inorganic hybrid materials. It consists of following steps:

Preparation of sol: Dissolve titanium isopropoxide (TTIP) in anhydrous isopropanol or ethanol with stirring. Use a typical molar concentration of 0.1 M TTIP.

Conversion of sol to gel: Deionized water and a catalyst (e.g., Nitric Acid, or Hydrochloric Acid, are added to TTIP under constant magnetic stirring to create sol. The reaction.

$$Ti(OC_3H_7)_4 + 4\,H_2O \rightarrow Ti(OH)_4 + 4C_3H_7OH \text{ (Hydrolysis)}$$

Here, titanium isopropoxide reacts with water forming titanium hydroxide and isopropanol.

$$2Ti(OH)_4 \rightarrow (OH)_3Ti\text{-}O\text{-}Ti\,(OH)_3 + H_2O \text{ (Condensation)}$$

The condensation of hydroxyl groups produces a three-dimensional Ti–O–Ti network with elimination of water, leading to formation of amorphous TiO_2.

iii) **Aging of gel:** Gel on aging for a known period of time which facilitates the formation of nanoclusters.

iv) **Removal of solvent:** The encapsulated liquid from gel can be removed by evaporation.

v) **Heat treatment:** Heating at 300°C to 500°C to achieve specific TiO_2 crystalline phases, such as anatase. Thin films produced via spin-coating allow for efficient resistive switching, enabling the creation of ReRAM devices.

Properties of TiO_2 for ReRAM

- **Electrical**: Shows resistive switching as oxygen vacancies move, creating or breaking conductive paths.
- **Structural**: Nanocrystalline anatase TiO_2 has many defects that improve switching and reliability.
- **Dielectric**: Wide bandgap (~3.2 eV) and high dielectric constant allow stable charge storage.
- **Scalability**: Easily integrates with CMOS, suitable for nanoscale devices.
- **Endurance & Retention**: Can last over 10^6 cycles and store data for more than 10 years.

Applications of TiO_2-based ReRAM

- **Non-volatile memory**: Faster and more energy-efficient than Flash memory.
- **Neuromorphic computing**: Works like artificial synapses for brain-inspired systems.
- **Logic-in-memory**: Combines storage and processing, cutting delay and power use.
- **Flexible & transparent devices**: Sol–gel TiO_2 films enable bendable, transparent electronics.
- **Next-gen storage**: Supports multiple resistance states for higher storage density.

Polymers

A polymer is a large molecule, or macromolecule, composed of many repeated subunits. The term "polymer" derived from the ancient Greek word *polus*, meaning "many, much" and *meros*, meaning "parts, and refers to a molecule whose structure is composed of multiple repeating units, from which originates a characteristic of highrelative molecular mass and attendant properties. The term was coined in 1833 by Jöns Jacob Berzelius. Becauseof their broad range of properties, both synthetic and natural polymers play an essential and ubiquitous role in everyday life. Polymers range from familiar synthetic plastics such as polystyrene to natural biopolymers such as DNA and proteins that are fundamental to biological structure and function. Polymers, both natural and synthetic, are created via polymerization of many small molecules, known as monomers. Their consequently large molecular mass relative to small molecule compounds produces unique physical properties, including toughness, viscoelasticity, and a tendency to form glasses and semicrystalline structures rather than crystals.

Definition of polymers: A polymer is a giant and macro molecule formed by the repeated union of several simple molecules. The repeat units in a polymer chain are linked through strong covalent bonds. The properties of polymers are entirely different from those of its monomers.

Monomers: A simple molecule having two or more bonding sites/functional groups through which can link to the other small molecules to form a polymer chain.

Degree of polymerization: This is a number which expresses the total number of monomers in a polymer chain.
Functionality: The total number of functional groups or bonding sites present in a monomer molecule is called the functionality of the monomer.

Polymerization: This is the process of conversion of monomers into polymers is known as polymerization.

Molecular Mass of Polymers

Molecular weight of any polymer is one of the important parameters as it links directly to the physical properties of polymers. In general, higher the molecular mass tougher is the polymer.
During the formation of polymers, different Polymers have different degrees of polymerisation, that is they have different chain lengths. The molecular Masses of individual macromolecules in a particular sample of polymer are different. Hence, an average value of the molecular mass is taken. there are two ways through which average molecular masses can be calculated

1. Number-average molecular mass
2. Mass- average molecular mass

Number-Average Molecular Mass

Number - Average molecular mass is the mass obtained when total mass of all the molecules of a sample is divided by the total number of molecules.
For example, in a particular sample, suppose
N_1 molecules have molecular mass M_1 each.
N_2 molecules have molecular mass M_2 each.
N_3 molecules have molecular mass M_3 each and so on.
Then, we have
Total mass of all the N_1 molecules $=N_1M_1$
Total mass of all the N_2 molecules$=N_2M_2$
Total mass of all the N_3 molecules$=N_3M_3$ and so on.
Therefore, Total mass of all the molecules $= N_1M_1+N_2M_2+N_3M_3= \Sigma N_iM_i$
Total number of all the molecules$=N_1+N_2+N_3=\Sigma N_i$
Hence the number-average molecular mass is given by:

$$\overline{Mn} = \frac{N_1M_1+N_2M_2+N_3M_3 +......}{N_1+N_2+N_3 +......}$$

$$\overline{M}_n = \frac{\sum N_iM_i}{\sum N_i}$$

Number - average molecular mass is determined by Osmotic pressure method.

Mass-Average Molecular Mass

Mass-average molecular mass is the mass obtained when the sum of the products of total mass of groups of molecules (having different molecular masses) and their respective molecular masses is divided by total mass of all the molecules.
For example, in a particular sample, suppose

N_1 molecules have molecular mass M_1 each.
N_2 molecules have molecular mass M_2 each.
N_3 molecules have molecular mass M_3 each and so on.
Then, we have
Total mass of all the N_1 molecules $=N_1M_1$
Total mass of all the N_2 molecules$=N_2M_2$
Total mass of all the N_3 molecules$=N_3M_3$ and so on.
The products of total mass of different groups of molecules with their respective molecular masses will be $(N_1M_1xM_1)$, $(N_2M_2xM_2)$, $(N_3M_3xM_3)$ et.
That is $N_1M_1^2$, $N_2M_2^2$, $N_3M_3^2$, etc.
Sum of products$=N_1M_1^2+N_2M_2^2+N_3M_3^2+...= \Sigma N_iM_i^2$
Therefore, Total mass of all the molecules $= N_1M_1+N_2M_2+N_3M_3= \Sigma N_iM_i$
Hence mass-average molecular mass is given by:

$$\overline{Mw} = \frac{N_1M_1^2+N_2M_2^2+N_3M_3^2 +......}{N_1M_1+N_2M_2+N_3M_3 +......}$$

$$\overline{M}_w = \frac{\sum N_iM_i^2}{\sum N_iM_i}$$

Mass-average molecular mass is determined by sedimentation

Distinguish between number average and weight average molecular mass of a polymer.

S. No	Number-Average Molecular Mass	Mass-Average Molecular Mass
1.	Total mass of all the molecules of a sample is divided by the total number of molecules.	The sum of the total mass of each individual polymers and their respective molecular masses is divided by total mass of all the molecules.
2.	$\overline{M}_n = \frac{\sum N_iM_i}{\sum N_i}$	$\overline{M}_w = \frac{\sum N_iM_i^2}{\sum N_iM_i}$
3.	Number-Average Molecular Mass is lower than Mass-Average Molecular Mass.	Number-Average Molecular Mass is higher than Mass-Average Molecular Mass.
4.	It is determined by Osmotic pressure.	It is determined by Sedimentation.

NUMERICAL PROBLEMS ON POLYMER

1. A polymer sample contains 2, 3 and 4 molecules having molecular weights 2×10^3, 3×10^3 and 4×10^3 respectively. Solve for the number average and weight average molecular weights of the polymer.

Answer: $\overline{Mn} = \frac{n_1 M_1 + n_2 M_2 + n_3 M_3}{n_1 + n_2 + n_3}$

$= \frac{2(2x10^3)+3(3x10^3)+4(4x10^4)}{2+3+4} =3.22 \times 10^3$

$\overline{Mw} = \frac{n_1 M_1^2 + n_2 M_2^2 + n_3 M^2_3}{n_1 M_1 + n_2 M_2 + n_3 M_3}$

$=2(2x10^3)^2+3(3x10^3)^2+4(4x10^4)^2 = 3.41x10^3$

$$(2x10^3)+3(3x10^3)+4(4x10^4)$$

2. A polymer sample has population as follows: 5 molecules of molecular mass each= 2000; 4 molecules ofmolecular mass each = 3000; 3 molecules of molecular mass each = 4000. Solve for its number average and weight average molecular weights.

Answer: $$\overline{Mn} = \frac{n_1 M_1 + n_2 M_2 + n_3 M_3}{n_1 + n_2 + n_3}$$

$$= \frac{5(2x10^3)+4(3x10^3)+3(4x10^3)}{5+4+3} = 2833.3$$

$$\overline{Mw} = \frac{5(2x10^3)^2+4(3x10^3)^2+3(4x10^3)^2}{5(2x10^3)+4(3x10^3)+3(4x10^3)} = 3058.82$$

3. Solve for M_n and M_W of a polymer which consists of 35% molecules having molecular mass 25000, 35% molecules having molecular mass 20000 and the remaining molecules having molecular mass 10000.

Answer: $$\overline{Mn} = \frac{n_1 M_1 + n_2 M_2 + n_3 M_3}{n_1 + n_2 + n_3}$$

$$= \frac{35x25000+35x20000+30x10000}{100} = 18750$$

$$\overline{Mw} = \frac{n_1 M_1^2 + n_2 M_2^2 + n_3 M^2_3}{n_1 M_1 + n_2 M_2 + n_3 M_3}$$

$$= \frac{35x(25000)^2+35x(20000)^2+30x(10000)^2}{35x25000+35x20000+30x10000} = 20733$$

4. Solve for number average and weight average molecular weights of a polymer consisting of 150 molecules with molecular mass 10^2g/mol, 200 molecules with molecular mass 10^3 g/moland 350 molecules with molecular mass 10^5g/mol.

Answer: $$Mn = \frac{n1\ M1 + n2\ M2 + ------}{n1 + n2 + -}$$

$$= \frac{150x10^2+200x10^3+350x10^5}{150+200+350}$$

$$=50307$$

$$\overline{Mw} = \frac{n_1 M_1^2 + n_2 M_2^2 + n_3 M^2_3}{n_1 M_1 + n_2 M_2 + n_3 M_3}$$

$$= \frac{150(10^2)^2+200(10^3)^2+350(10^5)^2}{150x10^2+200x10^3+350x10^5} = 99395.18$$

Structure – Property relationship

The structure of a polymer gives it specific physical and mechanical properties, which determine how useful it is for different applications. A polymer's properties depend on factors like its molecular arrangement, chain length, and how much cross-linking it has. Features such as its chemical makeup, crystallinity, and branching influence its strength, stability, and solubility. For example, more cross-linking makes a polymer stronger and more rigid, while higher crystallinity affects its mechanical behavior and how it deforms.

A. Tensile strength:

- Tensile strength is the maximum stress a polymer can withstand before breaking and depends on its molecular weight.
- It increases with degree of polymerization (DP) up to ~2000, after which the increase is minimal.
- Low molecular weight polymers are soft, while high molecular weight polymers are tough and heat resistant.

- Cross-linking improves strength (cross-linked > linear polymers).

B. Elasticity:

Elasticity is the ability of a polymer to stretch when a force is applied and to return to its original shape when the force is removed. This behavior occurs because the molecular chains uncoil during stretching and recoil when the force is released. Polymers exhibit elasticity as long as the chains do not separate from each other, preventing breakage during extended stretching. Elasticity can be enhanced by:

- Introducing cross-links at appropriate molecular positions to connect the chains,
- Avoiding bulky groups such as aromatic and cyclic structures in the repeating units, which hinder chain movement,
- Incorporating more non-polar groups in the chain to increase cohesion between chains, thereby reducing chain separation during stretching.

C. Chemical Resistivity:

- Chemical resistivity of a polymer refers to its ability to resist swelling, softening, dissolving, and loss of strength when exposed to solvents or chemicals.
- Polymers containing polar groups such as –OH or –COOH are typically dissolved by polar solvents like water or alcohols.
- Polymers with non-polar groups such as $–CH_3$ and C_6H_5 are not easily dissolved by polar solvents but dissolve readily in non-polar solvents like petrol, benzene, and carbon tetrachloride.
- Polymers with a higher degree of crystallinity exhibit greater chemical resistance than their amorphous counterparts.
- Chemical resistivity increases as the degree of cross-linking in the polymer structure increases.

D. Crystallinity

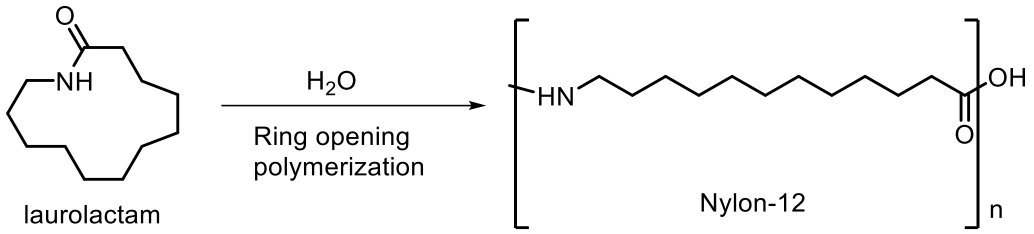
- Polymers contain both crystalline and amorphous regions, and crystallinity depends on chain structure (linear, branched, bulky groups).
- Linear, unbranched chains without bulky groups pack closely in an ordered manner, forming crystalline regions via intermolecular forces (Van der Waals, H-bonding).
- Higher crystallinity leads to increased tensile strength, impact resistance, density, and a sharp, high melting point.
- Examples of highly crystalline polymers include HDPE and stereoregular (isotactic and syndiotactic) polymers.

E. Plastic deformation

- Plastic deformation is the permanent change in shape of a polymer after removal of applied force, usually occurring near or above glass transition temperature (Tg).
- It happens due to sliding of polymer chains, especially in amorphous regions where chain mobility is higher.
- Amorphous polymers deform easily, while cross-linked polymers resist deformation due to restricted chain movement.
- It depends on factors like chain mobility, temperature, entanglement, and crystallinity, and is common in thermoplastics like polyethylene and PVC.

Nylon-12

Nylon 12 is a thermoplastic polymer produced from a monomer that contains 12 carbon atoms, such as laurolactam or ω-aminolauric acid. The polymerization process can be initiated by using a single monomer (laurolactam) or a co-polymerization of a diacid and a diamine with 12 carbons in their chains, such as 1,12-diaminododecane and 1,12-dodecanedicarboxylic acid.



Properties

- **Low water absorption**: Due to its long hydrocarbon chain, Nylon 12 absorbs very little water, which gives it high dimensional stability.
- **Mechanical strength**: It has high tensile strength, impact resistance, and excellent abrasion resistance, though lower impact resistance than some other nylons.
- **Chemical resistance**: It is resistant to a wide range of chemicals, including greases, oils, fuels, and hydraulic fluids.
- **Thermal properties**: Its melting point is around 176 °C (the lowest of all nylon polymers) and it has good thermal stability.
- **Other properties**: It has good dimensional stability, reasonable electrical properties, and is resistant to stress cracking.

Advantages in 3D printing applications

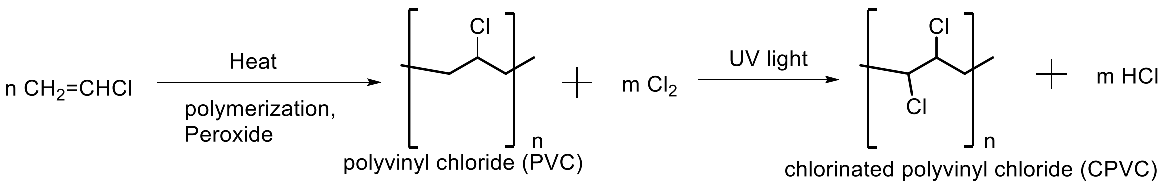
- **High Strength-to-Weight Ratio**: It provides a perfect balance of toughness and lightness. It is strong enough to replace metal in some automotive and aerospace applications, particularly when reinforced with carbon fiber.
- **Durability and Impact Resistance**: The material is malleable and can absorb significant impact without cracking. This makes it a top choice for functional moving parts, living hinges, and snap-fit enclosures.
- **Chemical and Thermal Resistance**: It is highly resistant to oils, fuels, and greases, which is critical for under-the-hood automotive parts. It also maintains stability in high-heat environments, with a heat deflection temperature around
- **Biocompatibility**: Many formulations meet for skin contact, allowing its use in custom prosthetics, orthotics, and dental aligners.
- Balances high-performance mechanical properties with exceptional reliability across multiple printing technologies

Chlorinated Polyvinyl Chloride (C-PVC)

CPVC is synthesized by further chlorinating polyvinyl chloride (PVC) . It is manufactured through methods like aqueous-suspension and processed using techniques similar to PVC. CPVC is produced by increasing the chlorine content of PVC, from about 56.8 wt% to 63–69 wt%.

Synthesis of CPVC is carried out in two main steps as follows:

Vinyl chloride monomers (VCM) undergo free-radical addition polymerization in the presence of a peroxide initiator under controlled heat and pressure to form polyvinyl chloride (PVC). The obtained PVC is exposed to chlorine gas under UV irradiation, during hydrogen atoms in the PVC chain are replaced by chlorine atoms, leading to the formation of chlorinated polyvinyl chloride (CPVC).

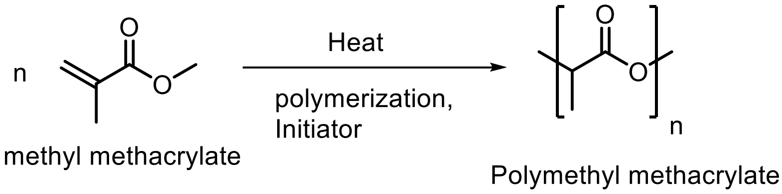


Properties of CPVC:

- CPVC is stronger and tougher than normal PVC.
- It has higher chlorine content, which makes it more rigid.
- CPVC can withstand higher temperatures (up to about 90–110 °C).
- It has good resistance to corrosion and many chemicals.
- CPVC is fire-retardant due to its high chlorine content.
- It shows good dimensional stability and does not warp easily.
- It is resistant to hot and cold water, so it is used in plumbing.
- CPVC has long service life and good mechanical strength.
- It can be easily processed by extrusion and molding.

Polymethyl methacrylate (PMMA) or Plexiglass

PMMA is synthesized by the free-radical bulk polymerization under an inert atmosphere using methyl methacrylate (MMA) monomers using initiators such as benzoyl peroxide or potassium persulfate and heat by addition polymerization.



Properties:

PMMA is a strong, tough, and lightweight material.
It is an amorphous and transparent plastic
It has good optical clarity but poor scratch resistance.
It has a density of 1.17–1.20 g/cm^3.
It also has good impact strength, higher than both glass and polystyrene, but significantly lower than polycarbonate and some engineered polymers.
PMMA ignites at 460 °C (860 °F) and burns, forming carbon dioxide, water, carbon monoxide, and low-molecular-weight compounds, including formaldehyde.
It is resistant to water, alkalis and inorganic salts but dissolves in organic solvents.
It has good resistance to acids and environmental deterioration.

PMMA in Device Applications

- **Optical devices**: Used in lenses, optical fibers, and light guides due to high transparency.
- **Display panels**: Utilized in LCD and LED screens for light transmission.
- **Medical devices**: Applied in intraocular lenses, bone cement, and dentures.
- **Solar cells**: As a transparent protective layer and encapsulant.
- **Microfluidic devices**: Due to easy machining and chemical resistance.
- **Automobile parts**: Headlight covers and indicator lamps because of clarity and weather resistance.
- **Protective shields**: Used as a glass substitute in safety windows and aquariums.

CONDUCTING POLYMERS

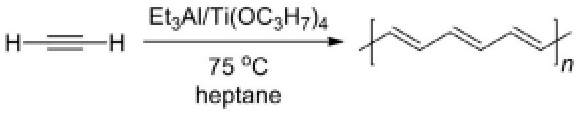
Conducting polymers or, more precisely, intrinsically conducting polymers (ICPs) are organic polymers that conduct electricity. Such compounds may have metallic conductivity or can be semiconductors. The biggestadvantage of conductive polymers is their process ability, mainly by dispersion. But, like insulating polymers, they are organic materials. They can offer high electrical conductivity but do not show similar mechanical properties to other commercially available polymers. The electrical properties can be fine-tuned using the methods of organic synthesis and by advanced dispersion techniques. Organic polymers with a highly delocalized pi electron system

having electrical conductance of the orderof a conductor is called conducting polymers.
Requirements for the polymer to be conducting:

1. Presence of conjugation throughout the molecule, which is given by delocalized pi and lone pair of electrons.
2. Linearity of the molecule

Synthesis of Polyacetylene

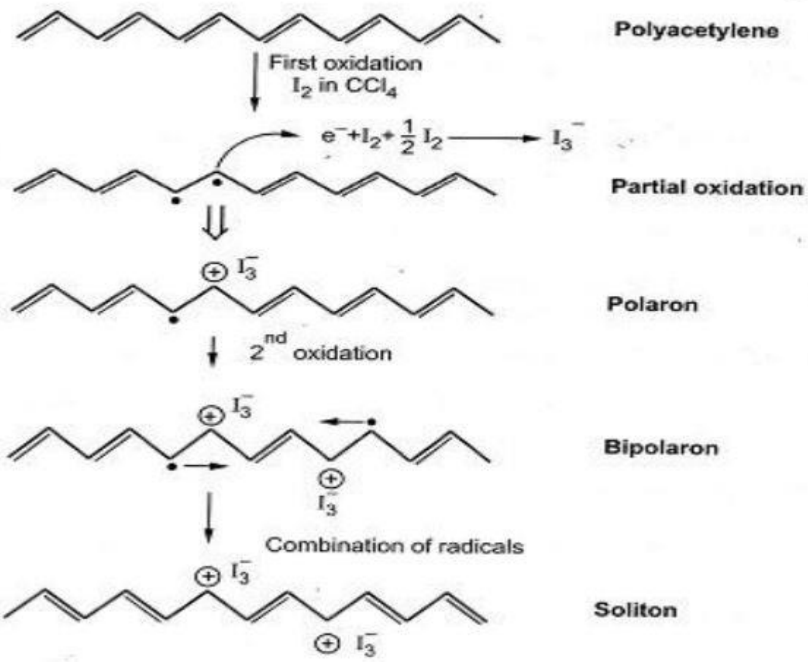
Polyacetylene is the simplest linear synthetic polymer. Fundamental unit consists from one carbon and one hydrogen atoms, as has been called simply (CH). Each carbon is σ bonded by sp^2 hybrid orbital to one hydrogen and two adjacent carbon atoms to form a planar chain molecule. Polyacetylene is synthesized by catalytic polymerization of acetylene using Ziegler-Natta catalyst $(Et)_3Al$ and titanium(IV) propoxide at 75°C, with a catalyst molar ratio (Al/Ti) of 2.5.



Synthesis of polyacetylene.

Conducting Mechanism in polyacetylene

Conductivity of pure pollyacetyline is 4.4×10^{-5} S/cm> upon doping with oxidizing agents like iodine conductivity increases to about 400S/cm.When the dopant iodine is added, it take away an electron from the π-backbone of the polyacetylene chain an creates a positive center(hole) on one of the carbon atoms. The other π-electron resides on other carbon atom making it a radical called polaron. Further oxidation leads to the formation of a bipolaron. Then the radicals migrate and combine to establish a backbone double bond. As the two electrons are removed, the chain will have twopositive centres (holes). The chain as awhole is neutral, but holes are mobile and when a potential is applied, they migrate from one carbon to another accounting for the conductivity as shown below.



Conducting Mechanism of Polyacetylene.

Before doping, there is sufficient energy gap between VB and CB so the electrons remain in VB and the polymer acts as an insulator. Upon doping, polarons and solitons are formed which results in the creation of new localized electronic states that fill the energy gap between VB and CB. When sufficient solitons are formed, a new mid-gap energy band is created which overlaps with conduction bands allowing electrons to flow.

Engineering applications:

1. **Conductive coatings** – Used in antistatic and electromagnetic shielding materials after doping.
2. **Organic electronics** – Acts as a base material for organic semiconductors in electronic devices.
3. **Energy storage** – Applied in rechargeable batteries due to reversible redox properties.
4. **Sensors** – Utilized in gas and chemical sensors because its conductivity changes with environment.
5. **Lightweight conductive materials** – Suitable for aerospace and advanced engineering applications due to low weight and conductivity.

Question bank

4-Mark Questions

1. What are semiconductors? Distinguish between conductors, semiconductors, and insulators.
2. Explain the concept of inorganic and organic semiconductors with examples.
3. Define p-type and n-type organic semiconductor with an example.
4. Write a short note on pentacene and perfluoropentacene as a semiconductor material.
5. List differences between organic and inorganic memory devices.
6. Explain the properties and applications of TiO_2 in memory devices.
7. Define number-average molecular mass (Mn)and weight-average molecular mass (Mw).
8. Distinguish between number-average molecular mass (Mn)and weight-average molecular mass (Mw).
9. Write properties of Nylon-12.
10. What are the requirements for a conducting polymer ?

6-Mark Questions

1. Explain the construction and working of pentacene semiconductor chip.
2. Describe the sol–gel synthesis of TiO_2 nanomaterials for ReRAM applications.
3. Explain structure–property relationship in polymers.
4. Explain properties and synthesis of Nylon-12 in 3D printing.
5. Describe the synthesis and properties of CPVC.
6. Explain the synthesis, properties, and applications of PMMA.
7. Explain the synthesis and mechanism of conduction in polyacetylene.